

Continuous Cognition, Failure-Atomic Actuation

A Protocol for Recovering Malformed Tool Calls Without Corrupting Stateful Agent Execution

Andrew Gracey

Independent Researcher GitHub: [plunder707](#)

Public artifact version 0.2.2, 2026-08-01

Abstract

Tool-using language systems combine probabilistic generation with state-changing operations. A malformed structured call is therefore not only a formatting error. If calls are parsed and dispatched sequentially, an early call can commit before a later call fails to parse. If the malformed response is also written into conversational history, recovery can begin from an inaccurate action trace. Existing work evaluates function-call correctness, structured-output validity, and final environment state. In the literature reviewed for this artifact, we did not identify an evaluation that isolates admission failure atomicity at the boundary between a generated action batch and its executor.

We define a protocol around one invariant: cognition remains continuous, tool admission is atomic, and side effects are committed individually. The protocol separates completed narrative content from action frames, requires a recognized terminal stop reason, validates every call in a batch before admitting any call, derives recovery language from an explicit admission record, and detects repeated malformed payloads by hash. The stop-reason gate is separate from validation because generation can stop on a valid structural boundary, leaving a batch that parses cleanly but is silently incomplete. It does not claim transactionality for arbitrary external effects. We motivate the design with a production incident in which output truncation produced an unterminated JSON argument and an uncaught exception after 30 successful tool executions. A source-faithful fault-injection harness reproduces the more general partial-batch counterexample. The sequential baseline executed the first action in the valid-then-malformed case and retained the malformed batch in history. Prevalidation reduced both outcomes to zero while preserving completed narrative content. Exhaustive truncation at 107 nonterminal byte positions produced the same contrast. A bounded behavior probe observed partial admission on all five pinned executable paths tested. A separately interpreted LlamaIndex typed core boundary rejected raw malformed arguments before an executable call could be represented; its provider-adapter behavior remains unresolved. The probe reports behavior on exact released paths. It does not establish that partial success is unintended, defective, or exploitable. These results demonstrate the invariant on the tested boundary harness and show its practical relevance through the incident and surface probes. A production-bound canary additionally exercised the deployed streaming adapter and admission policy with fixture-only rejection effects, while a live read-only request verified one admitted and one dispatched call. An experimental terminator-last control detected suffix loss but not silent middle-frame omission. These results do not establish end-to-end task improvement, exhaustive live-model recovery, or model-intent completeness.

1. Introduction

An action-generating system sits between two different semantics. Text generation is incremental and may stop at any token. External actions are discrete and may change files, databases, services, or remote systems. A runtime that treats a partially generated action batch as an ordinary assistant message allows the first semantic regime to leak into the second.

The failure is easy to miss because each component can behave as designed. The model endpoint reports an output limit. The streaming client preserves the fragments it received. The JSON parser rejects the incomplete string. The problem is the order in which the runtime admits and acts on those fragments.

The motivating incident occurred during a long-running implementation task. After 30 successful tool executions, the model emitted 931 characters of visible content, 795 characters of private reasoning, and one structured tool call. Generation stopped at exactly 12,000 tokens with `finish_reason=length`. The call's argument string ended inside a quoted value. The runtime had already appended the assistant response to history when an unconditional `json.loads` raised `JSONDecodeError`. The exception escaped the tool loop and terminated the turn. The model context was not exhausted, the tool-iteration ceiling was not reached, and the stack remained healthy.

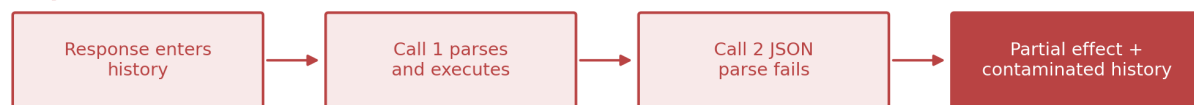
Increasing the generation limit reduces the frequency of this event but does not establish a correctness property. Any finite output can end during a structured call. The executor therefore needs an admission protocol that is correct under truncation, malformed completed output, and ambiguous execution acknowledgements.

We state the protocol as follows:

Cognition remains continuous; tool admission is atomic; side effects are committed individually.

Sequential admission leaks partial effects; batch prevalidation does not

Sequential baseline



Failure-atomic candidate



Invariant: cognition remains continuous; tool admission is atomic; side effects are committed individually.

Figure 1. Sequential admission can commit an early action before a later call fails. Failure-atomic admission buffers the action frame, validates the complete batch, and admits either the whole batch or none of it.

This wording separates three obligations.

- 1 Completed narrative content should survive rejection of an invalid action frame.
- 2 If any call in a proposed batch is invalid, no call from that batch should enter executable history or dispatch.
- 3 Once a valid call is dispatched, its effect cannot generally be rolled back. Recovery must report committed, failed, and unknown outcomes truthfully.

This paper makes six contributions.

- 1 It defines failure-atomic admission for generated tool-call batches and distinguishes it from rollback of external effects.

- 2 It gives a response-splitting and admission-record design that preserves completed cognition without retaining malformed action syntax.
- 3 It introduces bounded, state-derived recovery with repeated-payload detection.
- 4 It provides a source-bound fault-injection protocol, exhaustive byte-position truncation for a representative payload, and explicit claim gates for the end-to-end study.
- 5 It measures admission behavior on five pinned executable paths and one separately interpreted typed construction boundary.
- 6 It evaluates an optional terminator-last frame, showing that suffix loss is detectable in-band while silent middle-frame omission remains unresolved.

2. Problem Formulation

Let a generated response be

$$R = (C, F, A)$$

where C is completed narrative content, F is the finish reason, and $A = [a_1, \dots, a_n]$ is an ordered batch of proposed action frames. Each frame contains an identifier, a tool name, and a serialized argument object.

Let $V(a_i)$ be deterministic envelope and argument validation. Let $D(a_i)$ dispatch a validated action. Dispatch can produce one of three outcomes:

$$D(a_i) \text{ in } \{\text{committed}, \text{failed}, \text{unknown}\}.$$

The unknown state is necessary. A remote service can apply an operation and lose the acknowledgement, or a local process can time out after modifying its target. Retrying such an action automatically can duplicate the effect.

2.1 Admission failure atomicity

A batch satisfies admission failure atomicity when

$$\begin{aligned} & \text{NOT } \text{ALL}(i=1..n) \ V(a_i) \\ & \quad \text{implies} \\ & \text{for all } i,; \ a_i \text{ not in } H_A \text{ and } D(a_i) \text{ is not invoked,} \end{aligned}$$

where H_A is executable action history.

The property is deliberately narrower than a database transaction. It says that validation failure admits no action from the proposed batch. It says nothing about undoing calls that were admitted and dispatched successfully.

Validation is necessary but not sufficient. Generation can stop at the output limit on a valid structural boundary, between two complete calls rather than inside one. Every surviving frame then parses, no parser reports a fault, and a gate keyed only on $\text{bigwedge}_i V(a_i)$ admits a batch the model had not finished proposing. Completion is a property of the turn, not of the frames, so the admission condition requires a recognized terminal reason as well. Let T be the runtime's explicitly recognized terminal finish-reason set:

$$\text{admit}(A) \text{ iff } F \text{ in } T ; \text{and}; \text{ALL}(i=1..n) \ V(a_i).$$

This case is more dangerous than a parse failure because it is silent. A malformed frame raises. A cleanly cut batch executes a truncated plan and reports success.

2.2 Continuity preservation

If content C completed independently of the malformed action frame, the runtime may retain a content-only assistant message:

$$H_C \leftarrow H_C \ || \ C.$$

The malformed action bytes are excluded. This split prevents a formatting failure from erasing useful analysis while avoiding replay of broken syntax.

2.3 Threat and failure model

We consider accidental structural failures at the generation-execution boundary:

frame parses and the incomplete batch is indistinguishable from a finished one without the stop reason;

- output truncation inside an argument string;
- output truncation on a valid structural boundary, where every surviving
- an absent or unknown finish reason paired with fully parseable calls;
- syntactically invalid JSON with a non-length finish reason;
- missing action identifiers or tool names;
- valid JSON whose root is not an argument object;
- a malformed call before or after valid calls in the same batch;
- repeated emission of an identical malformed payload;
- failure after dispatch where commitment is uncertain.

Adversarial prompt injection, malicious tool implementations, authorization, and semantic correctness of valid arguments are outside the present claim.

2.4 Streaming boundary

Streaming does not change the admission invariant. Narrative content may be rendered incrementally because rendering does not authorize an external effect. Action frames require different treatment. Their deltas must remain buffered until the response terminates and the complete action batch has been validated. Dispatching a call as soon as its local JSON appears complete cannot establish batch failure atomicity because a later streamed call may be malformed.

The resulting constraint is:

Narrative may stream; executable action admission waits for terminal batch validation.

A runtime may expose provisional action deltas for observability, but those deltas cannot enter executable history or dispatch before the terminal admission decision.

2.5 Contract choice and scope

Failure-atomic admission is a proposed response-level contract, not the only possible tool-execution contract. Per-call independence is a defensible alternative contract: a runtime may execute each valid call and represent each invalid call as an isolated error. That design preserves useful partial work and is reasonable when calls are independent, idempotent, or read-only.

Response-level atomic admission is stronger. It is appropriate when calls are correlated parts of one plan, when any call can mutate external state, or when the runtime later replays the assistant response as one action proposal. The cost is that one malformed sibling suppresses otherwise valid work. A runtime can expose both policies, but the selected policy and its replay consequences should be explicit. This paper evaluates the stronger contract because the motivating runtime admitted state-changing calls and treated the response as one executable history entry.

3. Protocol

3.1 Split content from action frames

The runtime receives a response envelope but does not immediately append it to history. It first treats narrative content and action frames as separate admission candidates. Completed content can be retained without the action frames if action validation fails. Private reasoning is neither persisted nor returned as recovery context.

3.2 Check the stop reason, then prevalidate the complete batch

The first gate is the stop reason. If the turn did not end with a recognized terminal reason, the runtime rejects the whole assistant turn. Validation may still run to retain frame-level diagnostics, but it cannot authorize dispatch. A parse result cannot establish that the model finished speaking, and a truncated batch that happens to cut between calls parses cleanly. Salvaging the frames that parsed is exactly how a partial plan reaches dispatch.

Validation then checks every action before dispatch:

- 1 the call identifier is present;
- 2 the tool name is a non-empty string;
- 3 the argument payload is a string;
- 4 the payload parses as JSON;
- 5 the parsed root is an object;
- 6 optional tool-specific schema validation succeeds.

Only if the stop reason is recognized as terminal and all calls pass does the runtime append the full assistant tool-call message and begin ordered dispatch.

3.3 Record admission and execution separately

Each batch receives a record with:

- batch hash and width;
- finish reason;
- validation state;
- malformed call index and parser position, if any;
- admitted call count;
- per-call committed, failed, or unknown state;
- whether completed content was preserved;
- retry count and escalation.

This record is the authority for recovery wording. A static sentence such as "nothing executed" is unsafe because it is false after an ambiguous dispatch or for traces created before prevalidation existed.

3.4 Keep diagnostics out of cognitive replay

Operational telemetry stores bounded metadata:

- tool name;
- argument character count;
- error type and position;
- finish reason;
- batch width;
- payload hash;
- admission and execution states.

The malformed argument string is not placed back into the prompt. The retry instruction communicates semantics, not syntax:

The previous action batch was rejected before admission. No action from that batch was dispatched. Reissue a complete action batch if the operation is still required.

For an unknown execution state, the instruction changes:

An admitted action has unknown execution state. Do not repeat it automatically. Inspect the target system before deciding whether another action is safe.

3.5 Bound recovery without making small calls the default

The first structural failure receives one ordinary reissue opportunity. The runtime hashes the rejected batch. If the same hash fails again, it escalates to smaller complete calls. This ordering preserves the model's freedom to choose an appropriate operation in the common case while guaranteeing escape from an identical retry loop.

3.6 State machine

The protocol uses the following states:

```
received
-> rejected
-> validated
-> admitted
-> dispatched[i]
    -> committed[i]
    -> failed[i]
    -> unknown[i]
```

Only **validated** can transition to **admitted**. A rejected batch can preserve content, but it contributes no executable action frame to history.

Admission and execution use separate state transitions

Only a fully validated batch can cross the admission boundary.

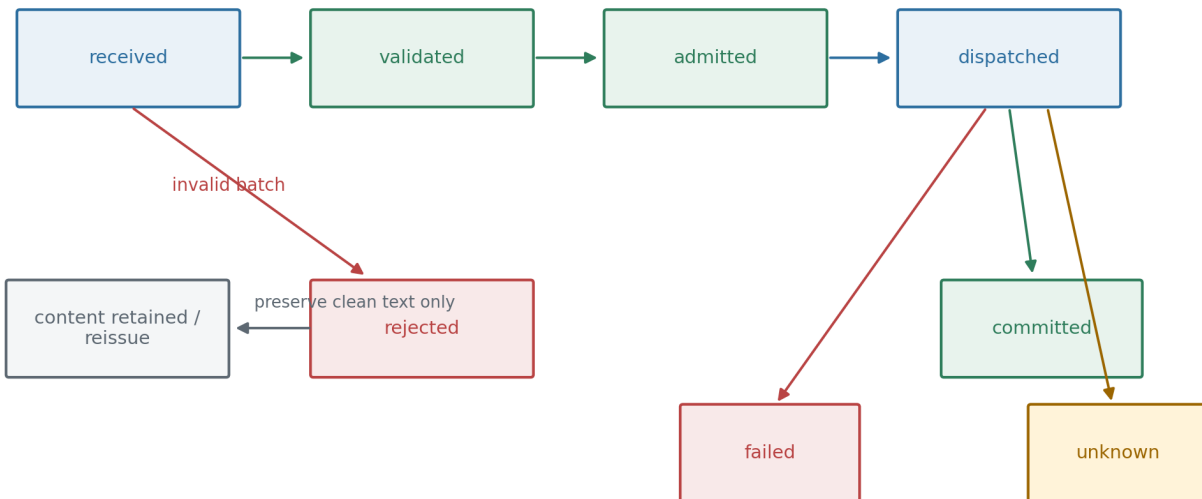


Figure 2. Admission and execution are separate state transitions. Unknown execution state is retained because acknowledgement loss can occur after an external effect commits.

4. Methodology

4.1 Evidence hierarchy

The study separates five evidence levels.

- 1 **Observed incident.** Live endpoint telemetry, server timing, traceback, process environment, and surviving filesystem artifacts establish the motivating failure.
- 2 **Source-backed mechanism.** The runtime source establishes that assistant history admission preceded per-call JSON parsing and dispatch.
- 3 **Deterministic counterexample.** A bounded harness reproduces the ordering with fake side effects and injected structural failures.
- 4 **Production-bound canary.** The deployed streaming adapter and admission policy must reject injected invalid batches before fixture effects and admit a valid terminal control batch.
- 5 **End-to-end evaluation.** A patched runtime must survive model-generated failures under long-horizon live replay. This stage remains incomplete.

The lower stages do not inherit authority from the higher ones. In particular, a design that passes the deterministic harness is not yet a production result, and a bounded production canary is not a complete recovery evaluation.

4.2 Fault-injection matrix

The deterministic case matrix covers:

and every frame parses;

- one malformed call with a length finish reason;

- valid then malformed calls;
- malformed then valid calls;
- two valid calls;
- completed content followed by a malformed call;
- scalar JSON arguments;
- a missing tool name;
- malformed arguments with a non-length finish reason;
- boundary truncation, where two complete calls carry a length finish reason
- an unknown finish reason paired with two complete calls;
- an acknowledgement loss after dispatch;
- an identical malformed retry.

Each case executes once. The harness is deterministic, and repeated identical runs would not add statistical information. In addition, one representative 108-byte argument is truncated at each of its 107 nonterminal byte positions. Every prefix is placed after one valid call, which tests whether fault position changes partial admission or history contamination.

The byte-position sweep cuts inside a single argument, so it cannot produce a boundary truncation. That limitation is why the case was absent from v0.1.0 and is stated here rather than left implicit. A sweep over multi-call payload positions would generate it directly and is the natural extension.

4.3 Bounded framework-surface behavior probe

The framework-surface artifact installs pinned releases in an isolated environment and injects one valid call followed by one truncated-JSON call. No network model is used. Each adapter records:

- whether the valid call produced a local effect;
- whether malformed action state remained represented;
- the exact released version;
- the execution surface and source-file digest;
- whether raw malformed arguments were representable at that boundary;
- the injection boundary and public, internal, or deprecated status;
- the state location and whether replay authority was established.

The tested surfaces are LangChain/LangGraph, AutoGen, LlamaIndex, the OpenAI Agents SDK, CrewAI, and smolagents. The adapters use each framework's released code, but not equivalent boundaries. Five adapters exercise executable paths. The LlamaIndex adapter exercises typed event construction because the raw malformed string cannot cross that boundary. The probes do not establish behavior for every provider, configuration, release, or language implementation.

LlamaIndex requires special interpretation. Its core `ToolCall` event requires already-parsed dictionary arguments, so the raw malformed call is rejected by the tested core type. Provider adapters perform parsing before this boundary. The core result is therefore structural rejection at one boundary, not proof that every LlamaIndex provider performs batch-wide prevalidation.

Table 1 records the comparability boundary that governs interpretation.

Project	Tested surface status	Injection boundary	Raw malformed input representable	Observed malformed-state location
--- --- --- ---	LangChain/LangGraph	Internal method of public component	Provider-normalized <code>AIMessage</code>	Yes, as <code>invalid_tool_calls</code>
	<code>AIMessage.invalid_tool_calls</code>	AutoGen	Public function	Model-client <code>FunctionCall</code> list
Yes	Generated <code>AssistantMessage</code>	OpenAI Agents SDK	Internal runtime path	

`ModelResponse.output` | Yes | Processed run items | | CrewAI | Deprecated internal method | Native call objects passed to handler | Yes | Executor assistant-message state | | smolagents | Internal method of public component | `Model-stub`
`ChatMessage.tool_calls` | Yes | `ActionStep.model_output_message` | | LlamaIndex | Public typed event |
`ToolCall` construction | No | None; construction rejected |

Only the first five rows form the executable-path denominator. The table also prevents state presence from being mistaken for equivalent replay authority: the exact state locations differ, and downstream replay was not established for every adapter.

4.4 Production-bound runtime canary

The production-bound receipt drives the deployed streaming adapter and response-level admission policy. Rejection cases dispatch only in-memory fixture functions. It tests a valid-prefix/malformed-sibling batch, two complete frames with `finish_reason=length`, and a complete terminal two-call control. A separate live request uses one read-only runtime diagnostic and compares the admitted-call and dispatched-call counters. Source snapshots and the upstream canary are bound by SHA-256. The private application source is not distributed, so this evidence is source-bound rather than independently executable here.

4.5 Metrics

We report:

- malformed-batch crash rate;
- partial-effect rate for valid-then-malformed batches;
- malformed-action history contamination rate;
- completed-content retention rate;
- truthful unknown-execution reporting rate;
- identical-retry escalation success;
- valid-batch regression;
- byte-position partial-effect and history-contamination counts;
- framework-surface partial-admission observations;
- production-canary admitted, dispatched, and fixture-effect counts;
- admission-decision latency for a valid two-call batch;
- terminator suffix-loss detection, false rejection, and middle-omission detection.

The end-to-end study will add recovery completion rate, additional model calls, latency, token overhead, task completion after recovery, and duplicate external effects.

4.6 Independent diagnosis

Agreement between systems that share the same framing is not independent evidence. A later diagnostic study will give each evaluator only the symptom, trace shape, and state transitions. It will omit the proposed invariant, terminology, and prior diagnoses. Evaluators will be isolated from one another's outputs. The study will measure whether they independently identify:

- batch prevalidation;
- content-action separation;
- state-derived recovery;
- ambiguous-execution handling;
- bounded retry.

This study tests diagnostic generality. It is not evidence for a general law of multi-agent intelligence.

4.7 Experimental terminator-last control

The supplementary deterministic arm appends a non-executable `__batch_complete__` frame after all proposed actions. Admission requires that frame to be last, requires its fixed versioned payload, then removes it before dispatch. This control detects a well-formed suffix cut even when the provider reports a terminal stop, because removing the suffix also removes the terminator.

The experiment includes two negative controls. A complete action batch that does not follow the terminator convention is rejected, measuring the protocol's false-rejection cost. A batch with one silently omitted interior action but an intact terminator is admitted, demonstrating that the marker establishes suffix completeness rather than model-intent completeness.

5. Results

The generated artifact is `artifact/results/fault_injection.json`. Its source binding includes the repository-relative path and SHA-256 digest of the sanitized reference ordering used to anchor the baseline.

The sequential baseline crashed on every malformed case and retained every malformed batch in history. In the valid-then-malformed case, it executed the first action before the second call failed to parse.

The candidate protocol rejected every malformed batch before admission. It executed no action in the valid-then-malformed case, retained completed narrative content in all content-plus-malformed cases, classified ambiguous post-dispatch failure as unknown, and escalated the second identical malformed payload to smaller complete calls. Valid two-call batches completed in order. It also rejected a fully parseable two-call batch carrying an unknown finish reason, while the sequential baseline executed both calls and reported completion.

Byte-position injection tested all 107 nonterminal cuts of the representative argument. The baseline executed the preceding valid action and contaminated history at all 107 positions. The candidate executed no action and retained no malformed action frame at all 107 positions.

The bounded behavior artifact is `artifact/results/framework_surface_probe.json`. All five pinned executable paths tested exhibited partial admission: LangChain/LangGraph 1.3.14/1.2.10, AutoGen 0.7.5, OpenAI Agents SDK 0.7.0, CrewAI 1.15.9, and smolagents 1.26.0. On each surface the valid call committed while malformed response state remained represented at the surface-specific location recorded in Table 1. The probe does not treat those locations as equivalent executable histories.

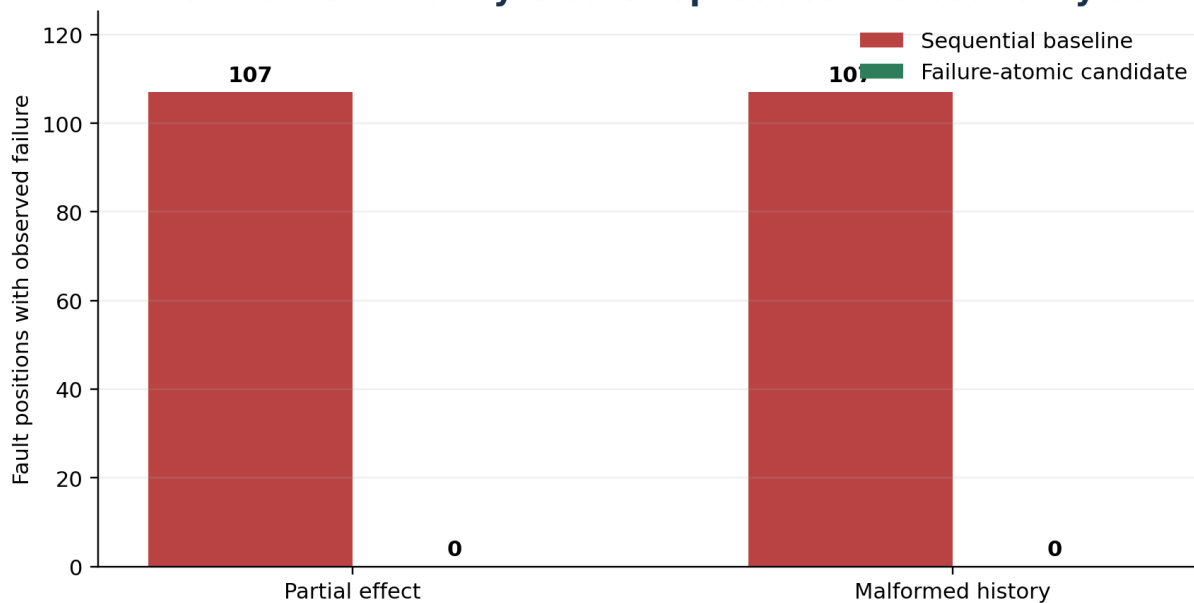
LlamaIndex core 0.14.23 rejected the malformed raw argument before constructing a `ToolCall` event. As noted above, this separately interpreted result leaves provider-adapter batch behavior unresolved and is not part of the executable denominator. The observed behavior does not by itself determine whether per-call partial success is intended framework policy, a defect, or an exploitable vulnerability.

The production-bound artifact is `artifact/results/production_admission_canary.json`. The deployed adapter and admission policy rejected both the mixed valid/malformed batch and the fully-parseable length-terminated batch with zero fixture effects. The valid terminal control admitted and executed two fixture calls. A separate live read-only request recorded one admitted and one dispatched call. The focused runtime regression suite passed 310 tests. A 20,000-iteration local microbenchmark measured approximately 31.5 microseconds per valid two-call admission decision.

The terminator-last artifact is `artifact/results/terminator_experiment.json`. The compliant two-action control completed without dispatching the terminator. The terminator gate rejected a terminal well-formed prefix that the stop-reason-only candidate admitted. It also rejected a complete but noncompliant batch. When the middle action was silently omitted but the terminator survived, the gate admitted the remaining action, preserving the stated intent-completeness limitation.

These results demonstrate the invariant on the tested boundary harness and show practical relevance through the incident, surface probes, and bounded production canary. They do not measure long-horizon recovery quality or the population frequency of malformed output.

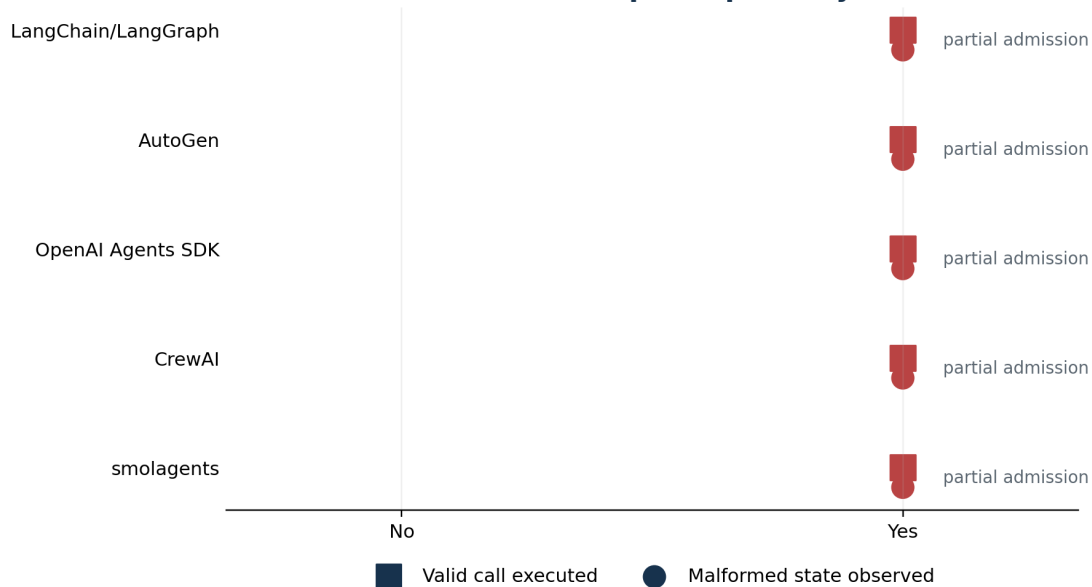
All 107 nonterminal byte cuts reproduce the boundary contrast



Representative argument: 108 bytes; every prefix followed one valid call.

Figure 3. Every nonterminal byte cut of the representative 108-byte argument produced a partial effect and malformed history in the sequential baseline. The candidate admitted neither.

All five tested executable paths partially admitted the mixed batch



Separate result: the tested LlamaIndex typed core boundary rejected raw malformed arguments.

Figure 4. All five tested executable paths partially admitted a mixed valid

and malformed batch. The separately tested LlamaIndex core boundary structurally rejected raw malformed arguments; its provider adapters remain unresolved.

6. Remaining End-to-End Evaluation

A complete evaluation needs five additional experiments.

6.1 Runtime streaming fault injection

Extend the three-case production canary to the completed byte-position matrix. Vary batch position, finish reason, tool type, and presence of completed narrative content. Assert that action deltas remain non-executable until terminal batch validation, then inspect history, telemetry, and side effects after each trial.

6.2 Live model replay

Replay long-horizon tasks under a fixed model, prompt, tool schema, and output budget. Compare the current runtime with the candidate on matched seeds. Measure task recovery rather than final prose alone.

6.3 Ambiguous external effects

Use a controllable service that can commit an operation and drop its acknowledgement. Verify that recovery never claims zero execution and never automatically repeats an unknown action.

6.4 Ablation

Remove one component at a time:

- content-action splitting;
- whole-batch validation;
- admission records;
- payload hashing;
- bounded escalation.

This identifies which component prevents each failure rather than attributing all gains to the complete protocol.

6.5 Performance

Extend the measured admission-decision latency to varying batch widths and payload sizes, then measure memory overhead, additional recovery calls, and prompt growth. The current two-call microbenchmark establishes only that the local validation cost is negligible relative to model and tool latency.

7. Related Work

Function-calling evaluations measure call selection, argument correctness, and multi-turn execution. BFCL extends function-call evaluation toward agentic settings [Patil et al., 2025]. ToolFailBench separates skipped tools, ignored results, fabricated outputs, and unnecessary calls [Soni, 2026]. These taxonomies concern whether tools are used correctly. Our focus is whether a proposed batch is admitted safely before any tool can be used.

Stateful benchmarks evaluate trajectories and world state. Tau-bench compares final database state with an annotated goal and reports reliability across repeated trials [Yao et al., 2024]. ToolSandbox evaluates intermediate milestones and forbidden minefields in stateful conversations [Lu et al., 2024]. E-Bench uses deterministic database-state differences for multi-step, state-changing tasks [Zheng et al., 2026]. Our protocol supplies a lower-level minefield: no state transition from a batch containing an invalid action frame.

Structured-output research addresses syntactic compliance. JSONSchemaBench evaluates constrained decoding over 10,000 schemas and shows that practical coverage, efficiency, and output quality must be measured separately [Geng et al., 2025]. Constrained decoding can reduce malformed calls, but it does not eliminate admission requirements. Output limits, transport faults, unsupported schema features, and non-constrained backends remain possible.

Isolation has also appeared at a different boundary. Repo2Run iteratively builds Docker environments and runs repository tests inside them [Hu et al., 2025]. Container isolation provides a stronger containment boundary for the commands it encloses. Our protocol applies before dispatch and remains useful when arbitrary external side effects cannot be contained or rolled back.

8. Limitations

The current evidence contains one production incident, a deterministic boundary harness, exhaustive truncation of one representative payload, five pinned executable-path probes, one separately interpreted typed-boundary probe, one three-case production-bound runtime canary, and one six-case experimental terminator control. All five executable paths admitted a valid call before handling a malformed peer, but this convenience sample is not a population-frequency estimate. It does not cover every framework release, configuration, provider adapter, language implementation, or custom execution loop. Several tested paths are internal methods, and the CrewAI result uses a released but deprecated executor path. The LlamaIndex result is limited to its typed core boundary and leaves provider normalization unresolved.

The probe does not establish that the observed behavior violates a documented framework contract. Per-call partial success and error isolation may be intentional. Nor does the presence of malformed state establish equivalent replay authority across frameworks. Maintainer interpretation and public end-to-end adapter tests are required before describing any individual result as a framework defect or security vulnerability.

The boundary harness mirrors the source ordering but is not the production runtime. The production canary covers one malformed-sibling batch, one fully-parseable length-terminated batch, and one valid control, but not the 107-position matrix or long-horizon model recovery. It does not evaluate semantic argument validity, authorization, malicious tools, concurrent dispatch, or distributed transactions. The production source is private and bound only by hashes in the public receipt, so independent reproduction of that canary requires access to the application runtime.

The terminator-last extension depends on model or provider compliance. It can fail loudly on a complete noncompliant batch, and it cannot detect a silently omitted middle frame when the terminator remains present. A trusted expected count or digest could detect some transport omissions, but it still would not prove which actions the model intended to generate. The released experiment therefore reports suffix completeness only and does not replace the stop-reason gate.

Content-action splitting assumes narrative content is independently complete. A model may refer to an action that was rejected, leaving the retained text misleading. A conservative implementation may need a content-retention policy or a visible marker that the proposed action was not admitted.

Batch prevalidation prevents partial effects caused by structural invalidity within the batch. It does not make a sequence of valid external actions transactional. If a later admitted call fails, earlier committed calls remain committed.

Finally, the terminology of cognition is functional, not ontological. Continuity here means preservation of completed task-relevant content and state, not a claim about consciousness or personhood.

9. Acknowledgements

The boundary-truncation case in Sections 2.1, 2.3, 3.2 and 4.2 was identified by a public reviewer after the v0.1.0 release, who observed that the stop reason and the parse result are different events and that keying admission on the parse alone leaves cleanly cut batches undetected. That correction is incorporated in v0.2.0. Follow-up public discussion about well-formed shortened lists motivated the terminator-last control in v0.2.2. The artifact preserves its false-rejection and middle-omission limits rather than treating the marker as an intent proof.

10. Artifact and Availability

The manuscript, deterministic harness, framework probes, exact result artifacts, figure-generation code, and tests are available at: <https://github.com/plunder707/failure-atomic-tool-admission>

The tagged release is the unit of citation. Code and executable artifacts are licensed under Apache License 2.0. The manuscript, figures, and documentation are licensed under Creative Commons Attribution 4.0 International.

11. Conclusion

For a correlated or mutating tool batch, malformed output should not be handled as ordinary conversational text or as the first failed command in a sequential batch. It is an admission failure for that response-level contract. The proposed boundary validates the complete batch, preserves completed content separately, records what was actually admitted and executed, and derives recovery from that record. Independent, idempotent calls may reasonably use the per-call contract discussed in Section 2.5.

The resulting invariant is narrow:

Cognition remains continuous; tool admission is atomic; side effects are committed individually.

Its value is that each clause can be tested, and none promises rollback the runtime cannot provide.

References

- 1 Patil, S. G. et al. "The Berkeley Function Calling Leaderboard (BFCL): From Tool Use to Agentic Evaluation of Large Language Models." ICML 2025. <<https://proceedings.mlr.press/v267/patil25a.html>>
- 2 Soni, H. "ToolFailBench: Diagnosing Tool-Use Failures in LLM Agents." arXiv:2607.04686, 2026. <<https://arxiv.org/abs/2607.04686>>
- 3 Yao, S. et al. "tau-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains." arXiv:2406.12045, 2024. <<https://arxiv.org/abs/2406.12045>>
- 4 Lu, J. et al. "ToolSandbox: A Stateful, Conversational, Interactive Evaluation Benchmark for LLM Tool Use Capabilities." arXiv:2408.04682, 2024. <<https://arxiv.org/abs/2408.04682>>
- 5 Zheng, W. et al. "E-Bench: Benchmarking Multi-Step Tool-Use Agents in Real-World Product Scenarios." arXiv:2607.23722, 2026. <<https://arxiv.org/abs/2607.23722>>
- 6 Geng, S. et al. "JSONSchemaBench: A Rigorous Benchmark of Structured Outputs for Language Models." arXiv:2501.10868, 2025. <<https://arxiv.org/abs/2501.10868>>
- 7 Hu, R. et al. "Repo2Run: Automated Building Executable Environment for Code Repository at Scale." arXiv:2502.13681, 2025. <<https://arxiv.org/abs/2502.13681>>